

Advay Kadam

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EDUCATION

University of Illinois Urbana-Champaign

Champaign, IL

B.S Computer Science and Statistics, Minor in Mathematics | GPA: 3.87

Expected Graduation: December 2026

- Relevant Courses: Algorithms and Models of Computation, Data Structures and Algorithms, Machine Learning, Computer Architecture, Time Series Analysis, Statistical Modeling, Statistics and Probability

TECHNICAL SKILLS

Languages: Python, C++, SQL, R, Java, JavaScript, HTML/CSS

Tools: PyTorch, Tensorflow, NumPy, Pandas, Flask, Azure, AWS SageMaker, Apache Spark/PySpark, MySQL, Docker

EXPERIENCE

Data Science Intern

June 2025 – Present

Bank of New York Mellon

Jersey City, NJ

- Summer 2025

Course Assistant for Data Structures and Algorithms

Jan 2025 – Present

University of Illinois Urbana-Champaign

Urbana, IL

- Leading office hours and discussion sessions on abstract data types, C++ algorithms, and complexity analysis
- Creating machine problems and auto-graded labs on graphs, trees, hash maps, arrays, and linked lists
- Developing and maintaining algorithmic exercises (e.g., nearest neighbor search) for 400+ enrolled students

Software Engineering Intern

May 2024 – Aug 2024

Procter & Gamble

Cincinnati, OH

- Optimized a sales ETL pipeline with PySpark/SQL using broadcast joins and partitioning to cut latency by 60%
- Boosted pipeline throughput 2.5x on 200k+ records/month using smart caching and DAG-aware joins in PySpark
- Implemented statistical anomaly detection using Isolation Forests with Python to flag deviations in batch loads
- Documented and diagrammed end-to-end ETL architecture to clarify pipeline flow and support future scalability
- Fine-tuned YOLOv9 model on retail image data to 90% accuracy and executed model on Azure ML pipeline

Undergraduate Researcher

Aug 2023 – May 2024

National Center for Supercomputing Applications

Urbana, IL

- Applied statistical analysis to real-time network data to assess post-quantum cryptographic algorithm resilience
- Analyzed sample of 2+ petabytes of network metadata using Python, Pandas, NumPy, and Shell scripting
- Identified that over 99% of SSH traffic lacked quantum-resistant encryption in collaboration with Dr. Phuong Cao
- Developed Flask dashboard to parse Zeek logs and deliver live metrics on PQC algorithm usage and vulnerability

Research Intern

Nov 2021 – Mar 2023

Rice University

Houston, TX

- Worked with Professor Aditya Mohite to enhance 2D-perovskite property predictions for photovoltaic applications
- Built PyTorch based regression model ($R^2 = 0.95$) to predict the energy-gap of 2D-halide perovskites
- Web-scraped and engineered 30+ chemical features using Pandas, NumPy, and Selenium for model training

LEADERSHIP/PROJECTS

Disruption Lab at Gies | *Lead Software Engineer*

- Led a team of 6 students to train transfer learning model to detect AI-generated deepfake images
- Created 2 custom datasets with 30k images and 140k images and trained model with Tensorflow and XceptionNet
- Deployed model to AWS SageMaker, boosting accuracy to 96% via automated hyperparameter tuning

Regression Library | *C++, Xtensor, Boost, CMake*

- Developed C++ library for multiple linear regression and hypothesis testing for reduced regression coefficients
- Implemented stochastic/batch gradient descent algorithms and confidence intervals for true parameter estimation

Time Series Regression Analysis of Sea Surface Temperatures | *R*

- Performed research on developing robust regression models for forecasting sea surface temperatures during El Niño
- Conducted residual diagnostics with ACF/PACF plots and Ljung-Box tests to validate autocorrelation structure
- Determined that regression on autocorrelated errors resulted in the best AIC (-2.7), R^2 (0.99), and RMSE (0.06)

ACHIEVEMENTS

- **IMC Trading Prosperity Competition** | Top 140/12,000 for Algorithmic Trading Challenge
- **Datathons** | 3rd place at UIUC Spring 2024 Datathon, 2nd place at UIUC Fall 2023 Datathon
- **Hackathons** | Best AI Project Prize at PioneerHacks III, Best Environmental Project Prize at WildHacks II